

1. Parse trees and derivations

1a.



$$E \rightarrow T \rightarrow F \rightarrow a$$

1b.



$$E \rightarrow E + T \rightarrow T + F \rightarrow F + a \rightarrow a + a$$

1c.



$$E \rightarrow E + T \rightarrow E + T + F \rightarrow T + F + a \rightarrow F + a + a \rightarrow a + a + a$$

1d.



$$E \rightarrow T \rightarrow F \rightarrow (E) \rightarrow (T) \rightarrow (F) \rightarrow ((E)) \rightarrow ((T)) \rightarrow ((F)) \rightarrow ((a))$$

2. Chomsky

Lets start by adding a new state A_0 and a rule $A_0 \rightarrow A$ we now have the states:

$A_0 \rightarrow A$
 $A \rightarrow BAB$
 $A \rightarrow 1$
 $A \rightarrow \varepsilon$
 $B \rightarrow 00$
 $B \rightarrow \varepsilon$

Now we remove $B \rightarrow \varepsilon$ and get:

$A_0 \rightarrow A$
 $A \rightarrow BAB$
 $A \rightarrow 1$
 $A \rightarrow \varepsilon$
 $A \rightarrow BA$
 $A \rightarrow AB$
 $A \rightarrow A$
 $B \rightarrow 00$

Now we remove $A \rightarrow \varepsilon$ and get:

$A_0 \rightarrow A$
 $A \rightarrow BAB$
 $A \rightarrow 1$
 $A \rightarrow BA$
 $A \rightarrow AB$
 $A \rightarrow A$
 $A \rightarrow BB$
 $A \rightarrow B$
 $B \rightarrow 00$

Now we eliminate $A \rightarrow A$ and get:

$A_0 \rightarrow A$
 $A \rightarrow BAB$
 $A \rightarrow 1$
 $A \rightarrow BA$
 $A \rightarrow AB$
 $A \rightarrow BB$
 $A \rightarrow B$
 $B \rightarrow 00$

Now we eliminate $A \rightarrow B$ and get:

$A_0 \rightarrow A$
 $A \rightarrow BAB$
 $A \rightarrow 1$
 $A \rightarrow BA$
 $A \rightarrow AB$
 $A \rightarrow BB$
 $A \rightarrow 00$
 $B \rightarrow 00$

Now we eliminate $A_0 \rightarrow A$ and get:

$A_0 \rightarrow BAB$
 $A_0 \rightarrow 1$
 $A_0 \rightarrow BA$
 $A_0 \rightarrow AB$
 $A_0 \rightarrow BB$
 $A_0 \rightarrow 00$
 $A \rightarrow BAB$
 $A \rightarrow 1$
 $A \rightarrow BA$
 $A \rightarrow AB$
 $A \rightarrow BB$
 $A \rightarrow 00$
 $B \rightarrow 00$

Now we eliminate all right hand variables longer than 2 and get:

$A_0 \rightarrow BX$
 $A_0 \rightarrow 1$
 $A_0 \rightarrow BA$
 $A_0 \rightarrow AB$
 $A_0 \rightarrow BB$
 $A_0 \rightarrow 00$
 $A \rightarrow BX$
 $A \rightarrow 1$
 $A \rightarrow BA$
 $A \rightarrow AB$
 $A \rightarrow BB$
 $A \rightarrow 00$
 $B \rightarrow 00$
 $X \rightarrow AB$

Finally we replace terminals longer than 1 with variables and get:

$A_0 \rightarrow BX$
 $A_0 \rightarrow 1$
 $A_0 \rightarrow BA$
 $A_0 \rightarrow AB$
 $A_0 \rightarrow BB$
 $A_0 \rightarrow 00$
 $A \rightarrow BX$
 $A \rightarrow 1$
 $A \rightarrow BA$
 $A \rightarrow AB$
 $A \rightarrow BB$
 $A \rightarrow YY$
 $B \rightarrow YY$
 $X \rightarrow AB$
 $Y \rightarrow 0$

Final form:

$A_0 \rightarrow BX \mid BA \mid AB \mid BB \mid YY \mid 1$
 $A \rightarrow BX \mid BA \mid AB \mid BB \mid YY \mid 1$
 $B \rightarrow YY$
 $X \rightarrow AB$
 $Y \rightarrow 0$

3. Show that i,j,k apples are context free

We can show that this is context free by creating a CFG that generates the language.

We start by creating a CFG for the left hand side apples, we want to be able to get infinite or none apples and finally end on a plus that leads into the right hand side. We'll say that $S \rightarrow aA$ and $A \rightarrow aA \mid B$

Now let's do the right hand side, it's similar to the left side, infinite or none apples but now we want the number of apples to match or be greater than the right side of $=$. We start by defining a new transition $S \rightarrow +aB$ and define $B \rightarrow aBa \mid aB \mid =$

We end up with the following CFG:

$$\begin{aligned} S &\rightarrow aA \\ S &\rightarrow +aB \\ A &\rightarrow aA \\ A &\rightarrow +B \\ B &\rightarrow aB \\ B &\rightarrow aBa \\ B &\rightarrow = \end{aligned}$$